DG Interns Hub – Week 1 Internship Report

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1. Executive Summary & Introduction

This comprehensive report details the rigorous theoretical and hands-on operational activities completed during Week 1 of the Cybersecurity Internship at DG Interns Hub.

Operating under Intern ID 008 from the National Forensic Sciences University (NFSU), the core objective was to bridge academic foundations with active cybersecurity methodologies by building an isolated home laboratory, mastering Linux command-line utilities, analyzing networking foundations, and completing gamified practical security challenges.

Key accomplishments throughout the week include deploying multi-virtual-machine configurations using Oracle VirtualBox, executing granular Linux user permission models, process inspections, and cron-based automation.

Furthermore, foundational knowledge was cemented through the successful completion of seven interactive TryHackMe rooms and clearing the first ten levels of OverTheWire's Bandit wargame.

This documentation has been structured to maintain high technical depth, absolute clarity, and zero plagiarism, serving as a comprehensive record of the week's execution.

2. Virtual Laboratory Architecture & Network Isolation

To ensure safe and controlled operational testing without exposing the host system to external threats, Oracle VirtualBox was deployed to provision three distinct virtual machines: a Kali Linux offensive testing workstation, an Ubuntu administrative server, and a Windows evaluation target.

Understanding network modes is critical for security professionals; NAT (Network Address Translation) permits guest VMs to access external networks while masking host identifiers, whereas Host-Only Networking creates a secure, isolated local subnet between the host and virtual machines without internet exposure. NAT Networks allow multiple internal VMs to communicate while sharing a singular gateway.

Crucial Security Precaution: Under no circumstances should vulnerable or intentionally compromised test environments (such as Metasploitable) be configured using Bridged Networking. Bridged mode bridges the virtual machine directly onto the physical local area network (LAN), exposing the vulnerable guest system to every device on the physical network and creating extreme security risks and unauthorized access vectors.

3. Linux System Administration & Privilege Management

Administrative operations were performed on the Ubuntu virtual machine to establish strict role-based segmentation and secure access control models.

Three dedicated user accounts (sec_analyst, pen_tester, and audit_guest) were established alongside two distinct security groups (blue_team and red_team) to mirror corporate multi-user environments.

Secure file collaboration was maintained by setting up a shared directory governed by restrictive file permissions using chmod (such as group-restricted mode 770) and assigning ownership using chown. Additionally, the security implications of the SUID (Set Owner Permission upon Execution) bit were evaluated, noting how misconfigured SUID binaries can lead directly to local privilege escalation.

System visibility and operational monitoring were maintained utilizing diagnostic utilities such as find for targeted file tracking, grep for pattern extraction in logs, ps aux for reviewing active processes, and ss -tulpn for auditing active listening network sockets and ports.

Routine system tasks and maintenance backups were successfully automated by configuring recurring jobs within the system cron scheduler (crontab -e).

4. Core Networking Foundations & Protocol Security

Network communication relies on foundational protocols: IP (Internet Protocol) manages logical packet routing across internetworks, MAC (Media Access Control) provides immutable hardware-level identification, DNS (Domain Name System) translates human-readable hostnames into numerical IP addresses, and DHCP (Dynamic Host Configuration Protocol) automates IP address assignment across local subnets.

Network reachability and diagnostic mapping across the home lab were verified using command-line tools: interface configurations were inspected via ip a, packet reachability was

tested with ping, packet routing paths were traced using traceroute, and DNS resolution queries were executed using nslookup and dig.

Port and service analysis is essential for both defensive hardening and offensive reconnaissance. Twelve core ports were analyzed, including Port 21 (FTP - cleartext credential exposure), Port 22 (SSH - brute-force susceptibility), Port 80/443 (HTTP/HTTPS - web application vulnerabilities), Port 3306 (MySQL - database exposure), and Port 3389 (RDP - remote desktop vulnerability vectors).

5. Gamified Practical Training: TryHackMe & OverTheWire

Practical skill development was reinforced by completing seven foundational TryHackMe modules: Search Skills, Learning Cyber Security, Linux Fundamentals Part 1, What is Networking, Introductory Networking, Networking Concepts, Pyramid Of Pain, and Intro to LAN, focusing on threat actor behavior, indicator categorization, and network packet flows.

Command-line mastery was rigorously tested by completing Bandit levels 0 through 10 on OverTheWire. Navigating plain-text SSH sessions required utilizing core Linux file examination tools—such as cat, file, base64, find, grep, sort, uniq, and strings—to locate hidden password keys across complex directory structures. In strict accordance with security guidelines, all level passwords have been omitted from public writeups while documenting the methodology thoroughly.

6. Theoretical Risk Frameworks & Professional Ethics

Information security governance is anchored by the CIA Triad (Confidentiality, Integrity, and Availability). Real-world defenses—such as multi-factor authentication, cryptographic hashing (e.g., SHA-256), and high-availability clustering—were mapped against each security pillar.

Clear distinctions were established across risk components: a Threat represents any potential danger or malicious actor, a Vulnerability is a software or architectural flaw, Risk measures the statistical probability of a threat exploiting a vulnerability, and an Exploit is the practical code or technique used to compromise the asset.

Clear operational boundaries separate SOC Analysts (continuous blue-team log monitoring and incident triage), VAPT Engineers (proactive red-team penetration testing and vulnerability assessments), and Incident Responders (forensic containment and remediation). All testing procedures strictly adhered to professional Rules of Engagement (RoE), formal authorization, and national cybercrime legislation.

7. Conclusion & Future Outlook

Week 1 successfully established an exceptionally rigorous foundation for Preethi R's cybersecurity internship at DG Interns Hub.

Through successful home lab deployments, advanced Linux administrative execution, network mapping, and gamified wargame completion, a high standard of technical capability was demonstrated.